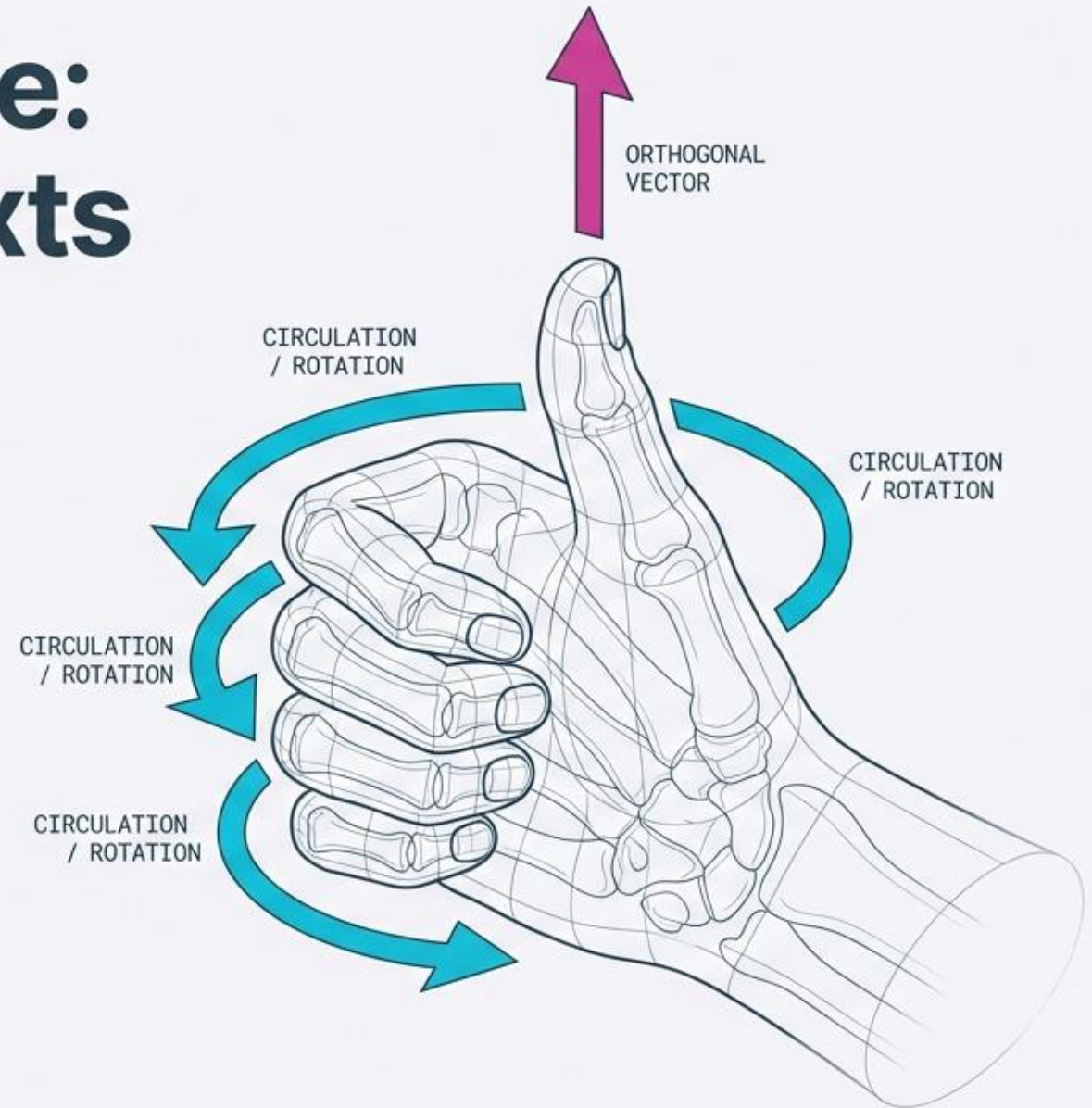


The Right Hand Rule: Two Critical Contexts

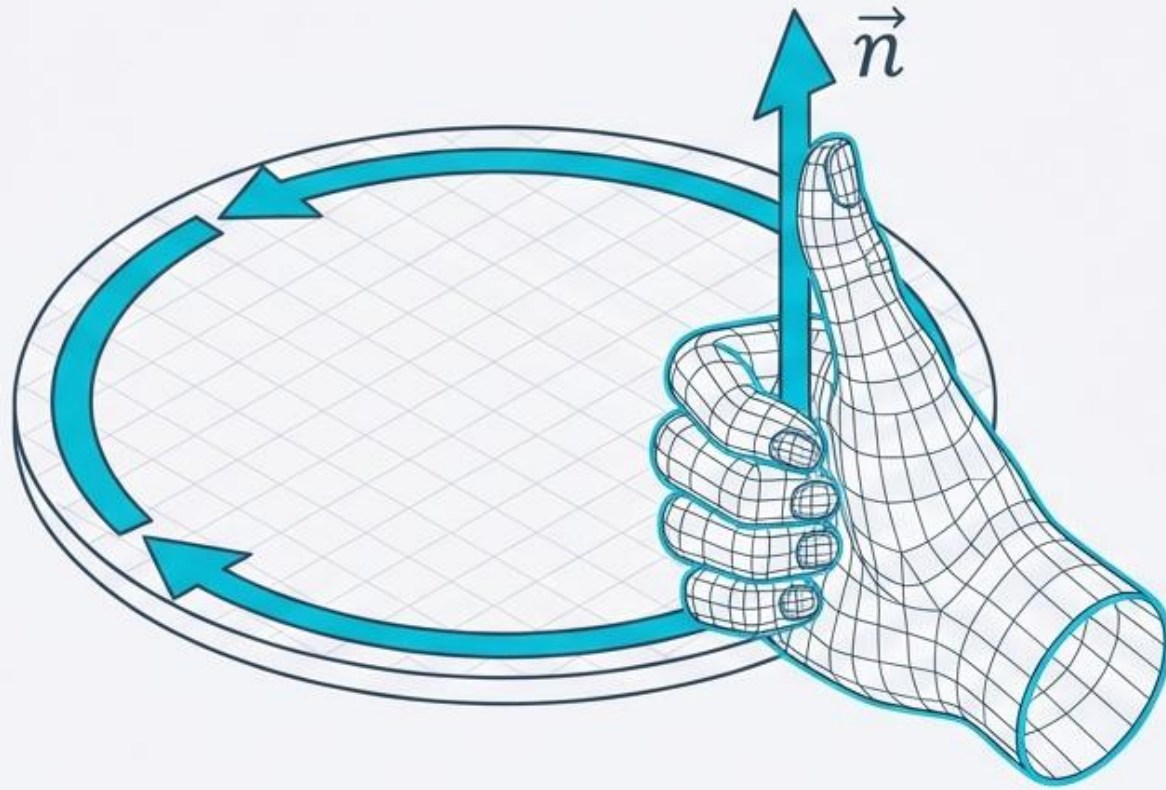
Disambiguating Geometric Definition from Physical Law

The Right Hand Rule connects rotational orientation to linear direction. To ensure precision in problem-solving, we must distinguish its use in defining spatial geometry from its application in electromagnetic induction.



Context I: Determining the Direction of vector $\vec{n}$

Geometric Definition (Agency)

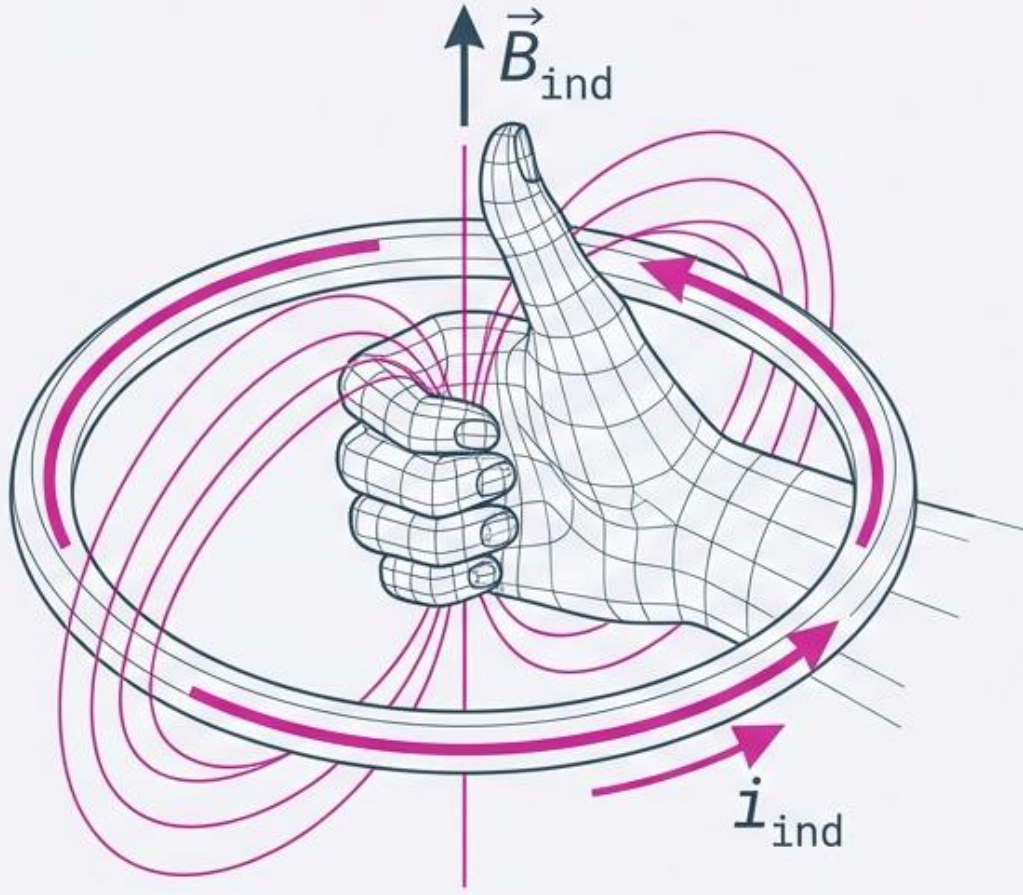


1. INPUT (The Choice): We choose a certain positive direction on the figure.
2. ACTION: Curl RH Fingers along this positive (+) direction.
3. OUTPUT (The Definition): The Thumb shows the direction in which the normal unit vector $\vec{n}$ points.

CRITICAL: In this context, the direction of vector $\vec{n}$ is a consequence of your choice of orientation.

Context II: Electromagnetic Induction

Physical Law (Causality)



1. INPUT (The Phenomenon): Identify the direction of the induced current (i_{ind}).
2. ACTION: Curl RH Fingers along the induced current direction.
3. OUTPUT (The Result): As a result, the Thumb points in the direction of the induced magnetic field (vector $\vec{B}_{ind}$).

CRITICAL: Here, the rule models a physical link: the current creates the field.

Functional Taxonomy: A Comparative Reference

Defining Vector $\vec{n}$	Determining Vector $\vec{B}_{\text{ind}}$
Trigger: We choose a positive direction. Relation: Orientation $\rightarrow$ Normal Vector	Trigger: Current i_{ind} flows. Relation: Current $\rightarrow$ Magnetic Field

Same Hand Motion.
Distinct Physical Meanings.